

## Time Evolution

In quantum mechanics, the time evolution of a closed system is governed by the time-dependent Schrödinger equation:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle. \quad (1)$$

For a time-independent Hamiltonian  $\hat{H}(t) = \hat{H}$ , we can readily solve this equation to find the time evolution of the state vector  $|\psi(t)\rangle$ :

$$|\psi(t)\rangle = e^{-\frac{i}{\hbar} \hat{H}t} |\psi(0)\rangle, \quad (2)$$

motivating us to define the time-evolution operator

$$\hat{U}(t) = e^{-\frac{i}{\hbar} \hat{H}t}. \quad (3)$$

If the Hamiltonian is time-dependent, the time evolution operator is given by the time-ordered exponential:

$$\hat{U}(t) = \mathcal{T} \exp \left( -\frac{i}{\hbar} \int_0^t \hat{H}(t') dt' \right), \quad (4)$$

where  $\mathcal{T}$  denotes the time-ordering operator. Time ordering is necessary because, in general, the Hamiltonian at different times may not commute,  $[\hat{H}(t_1), \hat{H}(t_2)] \neq 0$  for  $t_1 \neq t_2$ . So, the time evolution operator must account for the order in which the Hamiltonians are applied. If the Hamiltonian is time-dependent, but its evolution does not care about the order of things (i.e., if  $[\hat{H}(t_1), \hat{H}(t_2)] = 0$  for all  $t_1, t_2$ ), then time ordering is irrelevant and we get

$$\hat{U}(t) = \exp \left( -\frac{i}{\hbar} \int_0^t \hat{H}(t') dt' \right). \quad (5)$$



The result in Eq. (4) is given in S&N Chapter 2.1.2 and the derivation deferred to Chapter 5. You can find a sketch of the derivation in Section A, where we introduce the *Dyson Series*. The evaluation of time-ordered exponentials can be complicated and may require approximate methods to get to a solution. We will return to this issue in time-dependent perturbation theory, where the Dyson series will come back to aid (or bite?) us.

For a hermitian Hamiltonian, this operator is unitary, ensuring that the total probability is conserved over time. The state of the system at time  $t$  can be obtained by applying the time evolution operator to the initial state  $|\psi(0)\rangle$ :

$$|\psi(t)\rangle = \hat{U}(t) |\psi(0)\rangle. \quad (6)$$

The time evolution operator satisfies the following properties:

- Unitarity:  $\hat{U}^\dagger(t) \hat{U}(t) = \hat{U}(t) \hat{U}^\dagger(t) = \hat{I}$ .
- Composition:  $\hat{U}(t_2 + t_1) = \hat{U}(t_2) \hat{U}(t_1)$ .
- Initial condition:  $\hat{U}(0) = \hat{I}$ .

The expectation value of an observable  $\hat{A}$  at time  $t$  is given by

$$\langle \hat{A} \rangle_t = \langle \psi(0) | \hat{U}^\dagger(t) \hat{A} \hat{U}(t) | \psi(0) \rangle. \quad (7)$$

## Schrödinger versus Heisenberg Picture

From Eq. (7), we can see that there is an ambiguity in how we attribute the time dependence in quantum mechanics. One may easily interpret the expression as the time evolution operators acting on the states  $|\psi(0)\rangle$ , and then taking the expectation value of a fixed operator  $\hat{A}$ . But we may also equivalently interpret the expression as the time evolution operators acting on the operator  $\hat{A}$ , while the states remain fixed at their initial values. This leads to two different but equivalent pictures of quantum mechanics:

- **Schrödinger Picture:** States evolve in time, operators are fixed,

$$|\psi(t)\rangle = \hat{U}(t) |\psi(0)\rangle, \quad \hat{A}(t) = \hat{A}. \quad (8)$$

- **Heisenberg Picture:** Operators evolve in time, states are fixed,

$$|\psi(t)\rangle = |\psi(0)\rangle, \quad \hat{A}(t) = \hat{U}^\dagger(t) \hat{A} \hat{U}(t). \quad (9)$$

Which picture to choose from is often a matter of convenience. Schrödinger's picture is more intuitive when asking questions about the states themselves, such as what a wave packet does as it travels in space. Heisenberg's picture is often more convenient for dealing with time-independent states and focusing on the dynamics of observables, such as in time-dependent Hamiltonians. When studying time-dependent perturbation theory, we will make use of a hybrid version of these: the interaction picture. What is important is that the connection to physical observables, such as expectation values, remains the same in all prescriptions.



By differentiating Eq. (9), show that the Equation of motion in the Heisenberg picture is given by

$$\frac{d\hat{A}(t)}{dt} = \frac{1}{i\hbar} [\hat{A}(t), \hat{H}_H(t)] + \left( \frac{\partial \hat{A}}{\partial t} \right)_{\text{explicit}}, \quad (10)$$

where  $\hat{H}_H(t) = \hat{U}^\dagger(t) \hat{H} \hat{U}(t)$  is the Hamiltonian in the Heisenberg picture. Note that the additional term on the right-hand side is there to capture any explicit time dependence of  $\hat{A}$  (not common). If  $\hat{H}$  is time-independent, then  $\hat{H}_H(t) = \hat{H}$ . Note the important difference between  $\hat{H}_H$  and  $\hat{H}(t)$ : one is the Hamiltonian in the Heisenberg picture, the other is our original Hamiltonian as a function of time in (which we used assuming the Schrödinger picture, so you may also see  $\hat{H}(t) = \hat{H}_S(t)$ ).

## Propagators

The application of the time evolution operator can also be expressed in terms of *propagators*. These are Green's functions that describe how the wavefunction propagates from one point in space and time to another.



By a Green's function we mean a function that acts as the inverse of a differential operator, usually in the context of differential equations. For instance, the Green's function for the Laplacian operator  $\nabla^2$  is the function that satisfies  $\nabla^2 G(\mathbf{x}, \mathbf{x}') = \delta(\mathbf{x} - \mathbf{x}')$ , namely the function  $G(\mathbf{x}, \mathbf{x}') = -1/4\pi|\mathbf{x} - \mathbf{x}'|$ . As we will see, in quantum mechanics, propagators serve as Green's functions of the Schrödinger equation.

In the position basis, the propagator  $K(x_1, t_1; x_0, t_0)$  is defined as

$$K(x_1, t_1; x_0, t_0) = \langle x_1 | \hat{U}(t_1, t_0) | x_0 \rangle. \quad (11)$$

In the Schrödinger picture, we may be even as bold as to write

$$K(x_1, t_1; x_0, t_0) = \langle x_1, t_1 | x_0, t_0 \rangle, \quad (12)$$

where  $|x_1, t_1\rangle = \hat{U}(t_1, t_0) |x_1, t_0\rangle$ . Note what we have here: the propagator is the matrix element (aka, the transition amplitude, a type of expectation value) of the time evolution operator between two position eigenstates. In other words, we can say it is the amplitude for a particle to propagate from position  $x_0$  at time  $t_0$  to position  $x_1$  at time  $t_1$ .

How is this useful? Well, if we know the propagator, we can determine how any initial wavefunction  $\psi(x_0, t_0)$  evolves in time, and that is a big deal.



Specifically, show that the wavefunction at time  $t_1$  can be expressed in terms of the propagator and the initial wavefunction at time  $t_0$  as

$$\psi(x_1, t_1) = \int dx_0 K(x_1, t_1; x_0, t_0) \psi(x_0, t_0), \quad (13)$$

where  $\psi(x, t) = \langle x | \psi(t) \rangle$ . In addition, prove the following properties of the propagator:

- **Initial condition:** At equal times, the propagator reduces to the Dirac delta function

$$K(x_1, t; x_0, t) = \delta(x_1 - x_0). \quad (14)$$

- **Composition:** The propagator satisfies the composition property

$$K(x_2, t_2; x_0, t_0) = \int dx_1 K(x_2, t_2; x_1, t_1) K(x_1, t_1; x_0, t_0), \quad (15)$$

for any intermediate time  $t_1$ .

Great, if we know the propagator, we can find how any wavefunction evolves in time. But how do we find the propagator itself? It turns out that it obeys its own Schrödinger equation<sup>2</sup>:

$$i\hbar \frac{\partial}{\partial t_1} K(x_1, t_1; x_0, t_0) = \hat{H}(x_1, t_1) K(x_1, t_1; x_0, t_0), \quad (16)$$

with the initial condition  $K(x_1, t_1; x_0, t_1) = \delta(x_1 - x_0)$ . Solving this equation for the propagator can be challenging, but for simple systems like the free particle or the harmonic oscillator, explicit expressions for the propagator can be found. But most importantly, the propagator is a tool that is very useful to understand the path integral formulation of quantum mechanics and to solve many problems in quantum dynamics (as we will see later in this course). Ultimately, it is also a language that is fundamental to quantum field theory.

## Free Particle

Let us revisit the free particle Hamiltonian in three dimensions:

$$\hat{H} = \frac{\hat{p}^2}{2m} = -\frac{\hbar^2}{2m} \nabla^2. \quad (17)$$

The time evolution operator with  $t_0 = 0$  for the free particle is given by:

$$\hat{U}(t) = e^{-i\hat{H}t/\hbar} = e^{-\frac{i}{\hbar} \frac{\hat{p}^2}{2m} t}. \quad (18)$$

<sup>2</sup>You can show this by differentiating the definition of the propagator with respect to  $t_1$  and using  $\partial U(t_1, t_0)/\partial t_1 = -\frac{i}{\hbar} \hat{H}(t_1) U(t_1, t_0)$ .

To find the propagator in the position basis, we compute

$$K(\mathbf{x}_1, t; \mathbf{x}_0, 0) = \langle \mathbf{x}_1 | \hat{U}(t) | \mathbf{x}_0 \rangle = \langle \mathbf{x}_1 | e^{-\frac{i}{\hbar} \frac{\hat{p}^2}{2m} t} | \mathbf{x}_0 \rangle. \quad (19)$$

Inserting a complete set of momentum eigenstates, we have

$$K(\mathbf{x}_1, t; \mathbf{x}_0, 0) = \int d^3p \langle \mathbf{x}_1 | e^{-\frac{i}{\hbar} E_p t} | \mathbf{p} \rangle \langle \mathbf{p} | \mathbf{x}_0 \rangle, \quad (20)$$

where we defined the energy of the free particle as  $E_p = \frac{p^2}{2m}$ . Using the property  $\langle \mathbf{x} | \mathbf{p} = \frac{1}{(2\pi\hbar)^{3/2}} e^{\frac{i}{\hbar} \mathbf{p} \cdot \mathbf{x}}$ , we find

$$K(\mathbf{x}_1, t; \mathbf{x}_0, 0) = \frac{1}{(2\pi\hbar)^3} \int d^3p e^{-\frac{i}{\hbar} E_p t} e^{\frac{i}{\hbar} \mathbf{p} \cdot (\mathbf{x}_1 - \mathbf{x}_0)}. \quad (21)$$

This integral is a relatively straightforward Gaussian integral and can be evaluated to yield

$$K(\mathbf{x}_1, t; \mathbf{x}_0, 0) = \left( \frac{m}{2\pi i \hbar t} \right)^{3/2} \exp \left( \frac{im |\mathbf{x}_1 - \mathbf{x}_0|^2}{2\hbar t} \right). \quad (22)$$

This propagator describes how a free particle's wavefunction evolves over time, spreading out as it propagates through space.

As expected, Eq. (22) satisfies the initial condition (with  $t \rightarrow 0$ ,  $K(\mathbf{x}_1, t; \mathbf{x}_0, 0) \rightarrow \delta(\mathbf{x}_1 - \mathbf{x}_0)$ ) and composition properties of the propagator.



Fourier transform the propagator in Eq. (22) back to momentum space to show that

$$K(\mathbf{p}_1, t; \mathbf{p}_0, 0) = \langle \mathbf{p}_1 | \hat{U}(t) | \mathbf{p}_0 \rangle = \delta(\mathbf{p}_1 - \mathbf{p}_0) e^{-\frac{i}{\hbar} E_{p_0} t}. \quad (23)$$

This shows that in momentum space, the free particle propagator simply adds a phase factor to each momentum eigenstate, a consequence of the fact that momentum eigenstates are energy eigenstates for the free particle.

## A Time-Ordered Evolution

In what follows, we give a brief derivation of Eq. (4) (well... as brief as I could make it). The essence of the derivation is based on the fact that for an infinitesimal time step  $dt$ , the unitary time evolution is

$$\hat{U}(t + dt, t) = \hat{I} - \frac{i}{\hbar} \hat{H}(t) dt + \mathcal{O}(dt^2).$$

with  $H(t)$  evaluated at time  $t$ . We will not assume anything about  $\hat{H}(t)$  other than that it is hermitian. In particular, we do not assume that  $\hat{H}(t_1)\hat{H}(t_2) = \hat{H}(t_2)\hat{H}(t_1)$  for  $t_1 \neq t_2$  (that is, the Hamiltonians at different times do not necessarily commute).

Knowing the form of the infinitesimal time evolution operator, we can build up the full time evolution operator by composing many infinitesimal steps:

$$\hat{U}(t = t_N, t_0) = \hat{U}(t_N, t_{N-1}) \hat{U}(t_{N-1}, t_{N-2}) \cdots \hat{U}(t_1, t_0). \quad (24)$$

It is important that the earliest time evolutions appear on the rightmost side of this product. Using our infinitesimal form for  $U(t + dt, t)$  for uniform steps  $\Delta t = t_{k+1} - t_k$ , we have

$$\hat{U}(t = t_N, t_0) = \left( \hat{I} - \frac{i}{\hbar} \hat{H}(t_{N-1}) \Delta t + \dots \right) \cdots \left( \hat{I} - \frac{i}{\hbar} \hat{H}(t_1) \Delta t + \dots \right) \left( \hat{I} - \frac{i}{\hbar} \hat{H}(t_0) \Delta t + \dots \right), \quad (25)$$

Multiplying these out gives us terms of several orders in  $\Delta t$  (the lowest order being  $(\Delta t)^0$  and highest order being  $(\Delta t)^N$ ). Let us start collecting terms order by order and take the continuum limit  $N \rightarrow \infty$  (with  $N \times \Delta t$  fixed), also order by order.

- Order  $(\Delta t)^0$ : There is only one such term, which is simply the identity operator:

$$\hat{I} \xrightarrow{N \rightarrow \infty} \hat{I}. \quad (26)$$

- Order  $(\Delta t)^1$ : There are  $N$  such terms, each coming from picking one factor of  $-\frac{i}{\hbar} \hat{H}(t_k) \Delta t$  from one of the  $N$  factors in Eq. (25), and  $\hat{I}$  from all the others. Thus, the sum of all order  $(\Delta t)^1$  terms is

$$-\frac{i}{\hbar} \sum_{k=0}^{N-1} \hat{H}(t_k) \Delta t \xrightarrow{N \rightarrow \infty} -\frac{i}{\hbar} \int_{t_0}^t dt_1 \hat{H}(t_1). \quad (27)$$

For a given  $t$ , this is a simple integral from  $t_0$  to  $t$ .

- Order  $(\Delta t)^2$ : These come from multiplying two  $\Delta t$  factors in Eq. (25) as well as from the  $\Delta t^2$  terms in the expansion of each factor (omitted in Eq. (25)). There are  $N(N-1)/2$  of the former and  $N$  such terms of the latter type. Note that factors only ever appear at the same time or at decreasing time orders since the earliest time evolutions appear on the rightmost side of the product in Eq. (25). Thus, the sum of all order  $(\Delta t)^2$  terms is

$$\left( -\frac{i}{\hbar} \right)^2 \sum_{k_2 \geq k_1} \hat{H}(t_{k_2}) \hat{H}(t_{k_1}) (\Delta t)^2 \xrightarrow{N \rightarrow \infty} \left( -\frac{i}{\hbar} \right)^2 \int_{t_0}^t dt_2 \int_{t_0}^{t_2} dt_1 \hat{H}(t_2) \hat{H}(t_1), \quad (28)$$

where terms with  $k_1 = k_2$  come from the  $\Delta t^2$  terms inside a single  $\hat{U}(t_{k+1}, t_k)$  factor. Stare at this  $\sum \rightarrow \int$  correspondence until it makes sense. For a fixed time  $t$ , we are integrating over all possible triangular regions in the  $t_1$ - $t_2$  plane such that  $t_0 \leq t_1 \leq t_2 \leq t$ .

- Order  $(\Delta t)^n$ : The pattern established above continues for higher orders in  $\Delta t$  and we can write it in general as:

$$\left(-\frac{i}{\hbar}\right)^n \sum_{k_n \geq k_{n-1} \geq \dots \geq k_1} \hat{H}(t_{k_n}) \hat{H}(t_{k_{n-1}}) \dots \hat{H}(t_{k_1}) (\Delta t)^n \quad (29)$$

$$\xrightarrow{N \rightarrow \infty} \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^t dt_n \int_{t_0}^{t_n} dt_{n-1} \dots \int_{t_0}^{t_2} dt_1 \hat{H}(t_n) \hat{H}(t_{n-1}) \dots \hat{H}(t_1).$$

We can say that for a fixed time  $t$ , we are integrating over all possible  $n$ -dimensional “polytopes” in the space of  $t_1$ - $t_2$ - $\dots$ - $t_n$  defined by  $t_0 \leq t_1 \leq t_2 \leq \dots \leq t_n \leq t$ .

Victorious in our quest, let us now collect all orders to find the *Dyson series*:

$$\boxed{\begin{aligned} \hat{U}(t, t_0) = & \hat{I} + \left(-\frac{i}{\hbar}\right) \int_{t_0}^t dt_1 \hat{H}(t_1) + \left(-\frac{i}{\hbar}\right)^2 \int_{t_0}^t dt_2 \int_{t_0}^{t_2} dt_1 \hat{H}(t_2) \hat{H}(t_1) + \dots \\ & + \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^t dt_n \int_{t_0}^{t_n} dt_{n-1} \dots \int_{t_0}^{t_2} dt_1 \hat{H}(t_n) \hat{H}(t_{n-1}) \dots \hat{H}(t_1) + \dots \end{aligned}} \quad (30)$$

This looks quite daunting, so which is why we introduce an important piece of notation, the *time-ordering operator*  $\mathcal{T}$ . The definition of time ordering is formally very straightforward:

$$\mathcal{T} \{ \hat{H}(t_1) \hat{H}(t_2) \} = \begin{cases} \hat{H}(t_1) \hat{H}(t_2), & \text{if } t_1 > t_2, \\ \hat{H}(t_2) \hat{H}(t_1), & \text{if } t_2 > t_1. \end{cases} \quad (31)$$

The ordering of the limits in the integral is very awkward, but by applying the time-ordering operator, one may symmetrize each nested integral and let  $\mathcal{T}$  take care of the ordering for us:

$$\int_{t_0}^t dt_n \int_{t_0}^{t_n} dt_{n-1} \dots \int_{t_0}^{t_2} dt_1 \hat{H}(t_n) \dots \hat{H}(t_1) = \frac{1}{n!} \int_{t_0}^t dt_1 \dots \int_{t_0}^t dt_n \mathcal{T} \{ \hat{H}(t_1) \dots \hat{H}(t_n) \}. \quad (32)$$

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A simple example is  $n = 2$ . In that case, we have

$$\int_{t_0}^t dt_2 \int_{t_0}^{t_2} dt_1 \hat{H}(t_2) \hat{H}(t_1) = \frac{1}{2} \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 \mathcal{T} \{ \hat{H}(t_1) \hat{H}(t_2) \},$$

since the time-ordering operator ensures that when  $t_2 > t_1$ , the operators are swapped.

Substituting this into Eq. (30) gives

$$\hat{U}(t, t_0) = \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^t dt_1 \dots \int_{t_0}^t dt_n \mathcal{T} \{ \hat{H}(t_1) \dots \hat{H}(t_n) \} \quad (33)$$

Setting  $t_0 = 0$  recovers the form stated above.